co-op.care — Complete Technical Build Brief

For: Jacob Pielke / Cohesion Labs From: Blaine Warkentine, MD MPH Date: March 6, 2026

Jacob — Everything Changed This Week

Josh Emdur confirmed as Co-Founder & Medical Director on March 5. We've designed a complete product architecture including a cooperative-owned long-term care insurance product with full actuarial modeling. The website is built and rendering. The federal funding streams are identified and applications are in process. This document covers everything that needs to be built, in what order, and why.

The engagement model: co-working. I build frontend and product logic against your backend architecture using AI tools. You architect, I implement, we iterate.

1. THE PRODUCT (What We're Actually Selling)

Four Tiers — Each Builds on the Last

Tier 1: Comfort Card — FREE (or $9.95/mo) Free when you offer 1+ service to neighbors through the Time Bank. Includes: caregiver training library, burnout assessment (CII-12), Time Bank credits, monthly virtual support circle, care plan builder, resource navigator.

Why it matters technically: This is the network builder. Every free signup creates a node in the Time Bank graph. The services they offer become matchable supply. Their Time Bank activity data feeds the Age at Home insurance discount engine (4+ hrs/month = 36% insurance discount).

Tier 2: Care Plan — $59/mo Everything in Comfort + physician-reviewed care plan + Letter of Medical Necessity (HSA/FSA eligibility = 28-36% tax savings) + free dental/vision + quarterly physician reviews by Josh.

Why it matters technically: This tier generates the clinical data (CRI-14 longitudinal tracking, physician care plans as FHIR CarePlan resources, LMN documents) that makes members insurable at lower rates. The physician relationship converts them from unknown risk to known risk.

Tier 3: Shield — $149/mo Everything in Care Plan + access to professional caregivers at $35/hr (book by the hour, flexible week to week) + comprehensive dental/vision + caregiver career pathway investment. Shield membership = 24% discount on Age at Home insurance (continuous clinical data reduces actuarial uncertainty).

Why it matters technically: This is where CareOS scheduling, favorites/rating, EVV, clinical documentation, and Medicare billing all activate. The heaviest technical lift.

Tier 4: Age at Home Insurance — from $55/mo (age 45 with discounts) Cooperative-owned long-term care insurance. Covers professional home care up to $5,250/month when triggered (2+ ADL impairment certified by Josh). Launching 2027. Premiums 30-50% lower than traditional LTC insurance because of the blended care model.

Why it matters technically: Requires underwriting data pipeline, claims processing, premium management, and reinsurance integration. Later priority but architecture decisions now affect this.

The Blended Care Model (Critical to Everything)

This is the structural innovation. NOT all professional care. Professional caregivers handle clinical needs (bathing, mobility, medication, medical coordination) at $35/hr. Time Bank neighbors handle non-clinical needs (rides, meals, companionship, errands, housekeeping) at $0 cash cost.

A member needing 20 hrs/week total support:

- ~12 hrs professional (clinical) = $1,819/mo to insurance pool
- ~8 hrs neighbor Time Bank (non-clinical) = $0 to pool
- Total support: 20 hrs. Pool cost: $1,819 vs. $3,031 if all professional.

The Time Bank layer absorbs 20-40% of care volume at zero cost. This is the actuarial engine that makes Age at Home affordable. Every piece of Time Bank tech we build directly reduces insurance premiums.

The Favorites / Rating System (NOT "Same Person Every Visit")

Families build a care team of 3-6 favorite caregivers over time. After every visit: 1-5 star rating + optional note. Families choose from favorites based on availability and that week's needs. New caregivers introduced gradually — family approves before first visit.

This is more resilient than a single dedicated caregiver (no single point of failure), more scalable (better pool utilization), and more honest (one person can't always be there). CareOS tracks: % visits by favorites, rating trends, continuity score per family.

Flexible Week-to-Week Scaling

Care needs aren't static. Families book professional hours through CareOS by the hour. 6 hours this week. 25 next week after a hospital discharge. 2 the week after when she's recovering well. Request neighbor help through Time Bank anytime. Insurance premium doesn't change with utilization. Monthly benefit cap ($5,250) is fixed. Care plan adjusts quarterly (Josh) or as-needed.

2. WHAT EXISTS (Built by Blaine)

Component	Status	Lines
CII-12 (Caregiver Intensity Index)	Production React component	~800
CRI-14 (Care Readiness Index)	Production React component	~1,100
Website v11 (full product site)	Rendering, all features	~400
Care Needs Estimator	In website, blended model	Integrated
Neighbor signup + anonymous board	In website, seed data	Integrated
Co-op interest voting + progress bars	In website, seed data	Integrated
Age at Home pricing display	In website	Integrated
Membership tier display (expandable)	In website	Integrated
Family support tool (standalone)	Burnout check, resources, training, cost calc, community tabs	~240
Total production code		**~5,400+**

Backend committed but not integrated:

- Aidbox (Health Samurai / Pavel) — FHIR R4 native, PostgreSQL/JSONB. Pavel is a friend, likely free initially with advisor equity.

3. WHAT NEEDS TO BE BUILT — PRIORITY ORDER

P1: CMS ACCESS Infrastructure

Deadline: March 20 application. July 1 launch.

ACCESS is a 10-year CMS model paying outcome-aligned payments for technology-supported chronic care. Four clinical tracks: early cardiometabolic (hypertension, obesity, prediabetes), cardiometabolic (diabetes, CKD, CVD), musculoskeletal (chronic pain), behavioral health (depression, anxiety). This is our primary federal revenue stream.

What ACCESS requires us to build:

A. FHIR R4 Data Submission API

- Report clinical data through CMS-hosted APIs
- Patient demographics, condition-specific outcome measures, care plan documents

- Aidbox is FHIR R4 native — data model is aligned
- Office Hour March 18 3PM ET will publish API specs (I'm attending with Pavel)
- Build: API integration layer CareOS → CMS ACCESS endpoints
- Timeline: Architecture April, integration June, testing by July 1

B. Outcome Measurement Engine

- ACCESS pays based on % of patients meeting targets (50% threshold Year 1)
- Track-specific: BP reduction (10mmHg), HbA1c improvement, PHQ-9/GAD-7 scores, functional improvement
- CRI-14 captures baseline. Need longitudinal delta tracking + aggregated reporting
- Build: Dashboard monitoring per-member progress. Alerts for falling behind. Aggregate reporting for CMS.

C. HIE Connectivity

- Electronic care plan sharing at initiation, milestones, completion
- Direct messaging (Direct Protocol) minimum requirement
- PointClickCare/Collective Medical ADT for hospital discharge alerts
- Build: Direct messaging integration for care plan transmission to PCPs

D. Co-Management Billing Support

- PCPs reviewing ACCESS updates can bill ~$100/beneficiary/year
- Build: Automated care update generation from visit data → formatted for PCP review → trackable

P2: CareOS Core Platform

Needed for operations Q3 2026

A. Caregiver Favorites & Rating System

Data model:

- Member → has_many FavoriteCaregivers (refs, rating history, approval status)
- Visit → belongs_to Caregiver, Member
- Rating → score (1-5), note (optional), timestamp, visit_ref
- ContinuityScore → calculated field (% visits by favorites over rolling 90 days)

UI requirements:

- Family dashboard: favorites grid with photos, ratings, availability calendar, book-by-hour interface
- Caregiver dashboard: their families, upcoming schedule, ratings received, earnings
- Admin: matching suggestions, new caregiver introductions requiring family approval
- Reporting: continuity metrics per family, rating trends, caregiver quality scores

B. Visit Scheduling & Booking Engine

- Families book favorite caregivers by the hour, for specific days/times
- Real-time availability from caregiver schedule

- Flexible: different hours every week
- If favorite unavailable: suggest next-highest-rated from family's team
- Auto-reminders to family and caregiver
- Cancellation/rebooking flow
- EVV (Electronic Visit Verification) on every visit — GPS + time stamp (required for Medicaid and Class B)
- Build: Scheduling engine, availability management, conflict resolution, notification system, EVV

C. Clinical Documentation Pipeline

- Every visit generates a clinical note
- Caregiver enters natural language observations
- AI parses into Omaha System codes (42 Problem Classifications, 75 Intervention Targets)
- Omaha codes map to FHIR R4 resources (Observation, Condition, Procedure)
- FHIR resources route to multiple destinations:
  - Billing: CHI/PIN Medicare codes (G0019, G0022, G0023, G0024)
  - Reporting: ACCESS outcome tracking
  - Care planning: Josh's quarterly review queue
  - Insurance underwriting: CRI trend data
  - Quality: caregiver performance metrics
- "Capture once, route many" — one note, multiple output streams
- AI layer validated by Automate Clinic Reality.check (Josh is CMO)
- Build: NLP pipeline → Omaha classification → FHIR generation → multi-destination router

D. Care Plan Workflow Engine

- CRI-14 assessment → initial care plan recommendation (AI-generated)
- Blaine reviews and flags
- Josh reviews, edits, signs (digital signature on LMN)
- Signed care plan stored as FHIR CarePlan resource
- Quarterly reviews triggered automatically (or by: CRI delta > 15, hospitalization, fall, family request)
- LMN document generation: PDF with ICD-10 codes, CRI scores, service recommendations, physician signature, HSA/FSA qualification language
- Build: Generate → review queue → sign → store → monitor → trigger re-review workflow

E. CII-12 / CRI-14 Backend Integration

- Currently standalone React components — need persistence
- Aidbox FHIR Observation resources for each score
- Longitudinal tracking: score-over-time per member
- Alert thresholds: CRI > 75 = physician escalation. CRI delta > 15 in quarter = care plan review. CII > 65 = caregiver support intervention.
- Aggregate reporting for ACCESS outcomes
- Build: Assessment UI → Aidbox persistence → longitudinal dashboard → threshold alerting → outcome reporting

P3: Time Bank Platform

Critical — this is the actuarial engine for Age at Home insurance

A. Member Service Profiles

- Each neighbor lists services offered (12 categories: rides, meals, companionship, groceries, housekeeping, pet care, tech help, reading mail, yard work, repairs, respite sitting, exercise buddy)
- Availability preferences (days, times, frequency, max hrs/week)
- City/neighborhood for geographic matching
- Data model: User → has_many OfferedServices (category, availability, location, active status)

B. Request & Match Engine

- Families post care requests: "Need a ride Thursday 2pm" / "Need meals this week Mon-Fri"
- System matches nearby neighbors with matching services + availability
- Neighbor receives notification → accepts or declines
- If declined, next-best match notified
- Geographic proximity matching (zip code radius or neighborhood)
- Build: Request → match → notify → accept → confirm → complete → credit flow

C. Time Credit Ledger

- Double-entry system: every exchange creates credit (giver) and debit (receiver)
- Balance per member: hours given, hours received, net
- Credits transferable to family members (daughter earns hours, mother uses them)
- Monthly activity reports
- Insurance discount eligibility: 4+ hrs/month activity = 36% Age at Home discount (vs. 24% Shield-only)
- Build: Ledger system, balance tracking, transferability, activity reports, insurance eligibility engine

D. Anonymous Activity Feed

- Public-facing real-time feed: "A neighbor in Boulder is offering Rides, Companionship — 2 hours ago"
- No names until matched with a family
- Currently mocked with seed data in React website — needs real backend
- Updates as people join and complete exchanges
- Build: Event stream from Time Bank activity → anonymize → public WebSocket/SSE/polling feed

P4: Insurance Data Infrastructure

Needed for Age at Home launch 2027

A. Underwriting Data Pipeline Continuous member data feeds the actuarial model:

- CRI-14 scores (quarterly) — clinical trajectory
- CII-12 scores (quarterly) — caregiver capacity / burnout risk
- Visit frequency, type, and caregiver ratings
- Medication adherence tracking
- Fall history and prevention participation
- Hospitalization events (PointClickCare ADT alerts)
- Time Bank engagement (social isolation proxy — strong predictor of institutionalization)
- Wearable data (2027+ via HealthMCP/Nori — Apple Watch, Oura, Garmin, WHOOP)

Build: Analytics layer pulling from Aidbox FHIR store. Per-member risk scoring. Cohort analytics for actuarial review. This is the data moat — no traditional carrier has continuous longitudinal member health data.

B. Premium Calculation Engine Age at Home premiums depend on:

- Age at entry (45-65 bands)
- Tier: standalone vs. Shield discount (24%) vs. Shield + Time Bank discount (36%)
- Couple discount (additional ~20%)
- 3% compound inflation built in
- Build: Premium calculator + payment processing + discount eligibility verification (Is Shield active? Is Time Bank 4+ hrs/month?)

Pricing table (with Shield + Time Bank discounts):

Age	Individual	Couple (each)
45	$55/mo	$42/mo
50	$75/mo	$58/mo
55	$105/mo	$82/mo
60	$145/mo	$112/mo
65	$195/mo	$150/mo

C. Claims Processing (2027)

- Trigger: 2+ ADL impairment certified by Josh
- Benefit authorization workflow
- Professional care hours tracked against $5,250/month cap
- Neighbor Time Bank hours documented (not billed to pool but tracked)
- Claims adjudication (automated with manual override)
- Build: Claims workflow compliant with Colorado DOI regulations

P5: Integrations

A. PointClickCare / Collective Medical ADT

- Real-time Admit/Discharge/Transfer alerts from 1,300+ hospitals

- Member discharged from BCH → alert → care coordination trigger → caregiver notification → scheduling
- Josh is BCH hospitalist — need to confirm their ADT feed
- Build: ADT listener → member matching → workflow trigger → notification

B. Medicare Billing Engine

- CHI: G0019 (initial), G0022 (subsequent) — Community Health Integration
- PIN: G0023 (initial), G0024 (subsequent) — Principal Illness Navigation
- CCM: 99490, 99491 — Chronic Care Management
- ACCESS OAPs — new billing codes coming 2026
- Josh's NPI as supervising physician
- Build: Visit documentation → code selection → claim generation → submission → remittance tracking
- Requires CMS-855B enrollment (submitting this month)

C. Aidbox / FHIR R4 (Pavel)

- All clinical data as FHIR resources: Patient, Practitioner, CarePlan, Observation (CRI/CII), Encounter (visits), Condition, Goal (outcome targets), Schedule, Appointment
- Aidbox provides: REST API, FHIR search, subscriptions (triggers), bulk data export
- Build: Resource mapping for all CareOS data types. Subscription triggers for alerts. Bulk export for ACCESS and insurance reporting.

D. Payment Processing

- Stripe for membership payments + care service payments
- HSA/FSA compatible merchant category codes (critical — this is a selling point)
- Insurance premium collection (2027)
- Build: Stripe integration with HSA/FSA MCC configuration

E. CRM (HubSpot Free)

- Website forms currently local React state — need to connect to HubSpot
- Four pipelines: Co-op Interest, Family Waitlist, Caregiver Interest, Neighbor Network, White Paper Downloads
- Email sequences by pipeline
- Build: Form submission → HubSpot API. Can be simple initially.

F. Wearable Integration (2027+)

- HealthMCP / Model Context Protocol (Nori)
- Apple Watch, Oura, Garmin, WHOOP
- Continuous biometric data → CRI longitudinal enhancement → insurance underwriting
- Build: Later priority. Architecture should accommodate.

4. ARCHITECTURE

```
FAMILIES                          CAREGIVERS
NEIGHBORS
```

```
  │                              │                          │
  │ Book favorites               │ Accept visits            │
Offer services
  │ Rate caregivers              │ Document care            │
Accept requests
  │ Request neighbor help        │ View schedule            │
Earn/spend credits
  │ View care estimates          │ Track earnings           │
  │                              │                          │
  └──────────────────────────────┴──────────────────────────┘
              │                              │
              │                              │
     ┌────────▼──────────────┐       ┌──────▼──────────────┐
     │   CareOS Platform     │       │   Time Bank Engine  │
     │                       │       │                     │
     │ Scheduling/Booking    │       │ Service Profiles    │
     │ Favorites/Ratings     │       │ Request/Match       │
     │ Clinical Docs         │       │ Credit Ledger       │
     │ (NLP→Omaha→FHIR)      │       │ Activity Feed       │
     │ Care Plan Workflow    │       │ Insurance Discount  │
     │ CII/CRI Tracking      │       │ Eligibility         │
     │ Outcome Measurement   │       │                     │
     │ EVV                   │       │                     │
     └───────────────────────┘       └─────────────────────┘
              │                              │
              │                              │
     ┌────────▼──────────────────────────────▼──────────────┐
     │              Aidbox / FHIR R4 (Pavel)                │
     │ Patient, CarePlan, Observation, Encounter,           │
     │ Condition, Goal, Schedule, Appointment               │
     │ + Time Bank Ledger (custom or FHIR extensions)       │
     └──────────────────────────────────────────────────────┘
              │                              │
              │                              │
     ┌────────▼──────────────┐       ┌──────▼──────────────┐
     │  Integrations         │       │ Insurance Layer     │
     │                       │       │ (2027)              │
     │ CMS ACCESS API        │       │                     │
     │ Medicare Billing      │       │ Underwriting Data   │
     │ PointClickCare ADT    │       │ Premium Engine      │
     │ HubSpot CRM           │       │ Claims Processing   │
     │ Stripe Payments       │       │ Reinsurance API     │
     │ Direct Messaging      │       │ Actuarial Reports   │
```

5. TECH DECISIONS NEEDED FROM YOU

Decision	Options	My Lean	Notes
Backend API	Node/Express, Python/	Your call	Must talk Aidbox FHIR API
Auth	Firebase Auth, Auth0,	Need	Multi-role: family, caregiver, neighbor,
Hosting	AWS, GCP, Azure	Need HIPAA	Backend must be HIPAA compliant
Frontend	Vercel, Netlify	Vercel	Deploys from GitHub, free tier
Database beyond FHIR	Postgres alongside Aidbox?	Your call	Time Bank ledger, scheduling, billing may need relational
Real-time	WebSockets, SSE,	Your call	For Time Bank feed + scheduling
NLP pipeline	Claude API, local model, both	Claude API initially	Reality.check validates; can move local later
Payment	Stripe	Stripe	Need HSA/FSA merchant codes
EVV	Custom GPS+timestamp	Custom	Required for Class B
Queue/jobs	Bull, Celery, SQS	Your call	For async: billing, notifications, ADT

6. DIVISION OF WORK

I build (with AI tools):

- All React frontend: website, dashboards, assessment UIs, booking interfaces, Time Bank UI
- CRI-14 / CII-12 clinical logic and scoring algorithms
- Care plan document generation (LMN PDFs, care summaries)
- Omaha System mapping logic (clinical taxonomy)
- Product design, UX, flows
- Content, copy, member-facing communications
- Clinical documentation templates
- Test data and scenarios

You architect and build:

- Backend API layer (routes, controllers, business logic)
- Aidbox/FHIR integration and resource mapping
- Database schema for non-FHIR data (Time Bank ledger, scheduling, billing, ratings)
- Authentication system (multi-role, HIPAA-compliant)
- CMS ACCESS API integration (after March 18 specs)
- PointClickCare ADT listener
- Medicare billing submission pipeline

- Stripe payment integration (including HSA/FSA merchant config)
- NLP pipeline infrastructure (I provide the prompt engineering, you provide the plumbing)
- Real-time infrastructure (Time Bank feed, scheduling updates, notifications)
- DevOps: CI/CD, HIPAA hosting, monitoring, backups
- EVV system
- Multi-tenancy architecture for federation (when ready)

7. TIMELINE

When	Build	Why
March	Aidbox instance (Pavel). CRI/CII persistence. Website deploy (Vercel).	Foundation
March 18	ACCESS Office Hour — get API specs	Critical deadline
March 20	ACCESS application submitted	Revenue stream
April	ACCESS data architecture. FHIR resource mapping. Auth system.	ACCESS
May	Scheduling/booking engine. Favorites/rating system. EVV. Basic care	Operations
June	Time Bank MVP (profiles, matching, credits, feed). Medicare billing engine. Direct messaging.	Launch prep
July 1	ACCESS Cohort 1 live. CareOS operational. First paid visits.	Revenue
Q3	Outcome measurement. PointClickCare ADT. Clinical documentation	Scale
Q4	Insurance data warehouse. Premium engine. Underwriting pipeline.	Age at Home prep
2027	Claims processing. Wearable integration. Multi-tenancy for federation.	Insurance

8. THE MARGIN ARCHITECTURE (So You Understand the Business)

Six layers protect the insurance pool — and several are technical:

1. **Time Bank absorbs 20-40% of care hours at $0.** Every Time Bank match we facilitate is money NOT coming out of the insurance pool. The better the matching engine, the lower the claims.

2. **Shield data reduces actuarial uncertainty 30-40%.** The continuous CRI/CII data pipeline is what justifies the 24% insurance discount. Better data infrastructure = more accurate underwriting = lower premiums = more members.

3. **Prevention reduces claims.** Every CRI alert that triggers early intervention, every fall prevented by training, every medication interaction caught by documentation = a smaller future claim or no claim.

4. **15% reserve margin in premiums.** Straightforward actuarial cushion.

5. **Cooperative controls care cost inflation at 3%.** Caregiver wages set by cooperative governance, not market bidding. Tech platform (what you're building) keeps admin costs low.

6. **No shareholder extraction.** 80% loss ratio target vs. traditional 60%. Members own the surplus.

9. THE PEOPLE

- **Blaine Warkentine, MD MPH** — Founder. Fellowship-trained orthopedic surgeon. 20 years healthtech. 5 patents. Multiple exits. Building frontend and product with AI tools. blaine@co-op.care.

- **Josh Emdur, DO** — Co-Founder & Medical Director. Confirmed March 5. BCH hospitalist since 2008. CMO Automate Clinic (Reality.check AI validation platform). Former CMO SteadyMD. Licensed in all 50 states. Handles clinical sign-offs (~1-2 hrs/month). The physician signature that makes HSA/FSA, insurance, and ACCESS all work.

- **Pavel (Health Samurai / Aidbox)** — Infrastructure partner. FHIR R4 native platform. Friend of Blaine. Likely free initially with advisor equity arrangement.

- **Jacob Pielke (Cohesion Labs)** — That's you. Backend architecture. Co-working model.

10. IMMEDIATE ASK

1. **Review this architecture.** What's missing? What's wrong? What would you change?
2. **Aidbox coordination with Pavel.** Can we get a dev instance this month?
3. **Auth recommendation.** HIPAA-compliant, multi-role (family, caregiver, neighbor, physician, admin). What do you suggest?
4. **Hosting recommendation.** HIPAA BAA required. Where?
5. **ACCESS API prep.** After March 18 Office Hour, I'll have specs. How fast can we start the integration architecture?
6. **Standing weekly sync.** 30 minutes. I come with prioritized tickets from this document.
7. **Website deployment.** Can we get the React site on Vercel this week? Just the frontend — forms stay local for now, connect HubSpot later.

11. WHY THIS MATTERS

This is not a startup building a product. This is a community building infrastructure.

The CMS ACCESS application is due March 20. The model launches July 1. MAHA ELEVATE is a $100M federal program with NOFO coming. The actuarial model for Age at Home shows premiums 30-50% below traditional LTC insurance. There are already 7 cities showing interest in co-op formation on the website. The neighbor network concept has people signing up for free. Josh Emdur — a 50-state licensed physician who's been at Boulder Community Health for 18 years — co-founded this because he sees what's coming.

70% of Americans over 65 will need long-term care. Only 11% have insurance. Nursing homes cost $10,965/month. The caregiver ratio drops from 7:1 to 4:1 by 2030. The professional system cannot scale.

But communities can. That's what we're building the technology for.

Let's talk.

— Blaine blaine@co-op.care